

ENGR 1451/2451: Exploratory Data Science

Correlation, Confounding, and Causation

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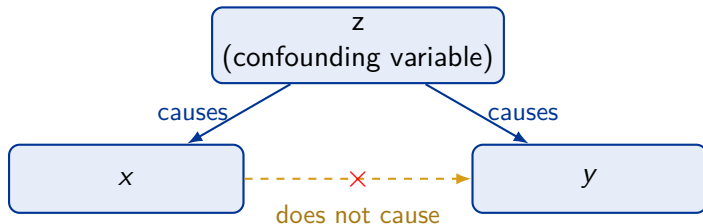
Today: Association and Causation

Correlation and regression describe how variables vary together. This lecture examines when a relationship supports a **causal claim**.

- Distinguish **association** from **cause and effect**.
- Compare **controlled experiments** with **observational studies**.
- Identify **confounding** and explain how randomization and careful comparisons help.
- Use **Simpson's paradox** to understand why overall and subgroup comparisons can differ.

Correlation does not imply causation

- Correlation measures association.
- However, association does not necessarily show causation.
- Both variables may be simultaneously influenced by some third variable.
- That third variable is known as a **confounding variable**



Ice Cream Sales and Drownings

Observed correlation

In many places, monthly **ice cream sales** and **deaths by drowning** tend to rise and fall together.

Consider the explanations

What are at least **two** possible explanations for this correlation?

A tempting claim

More ice cream consumption causes more drownings.

Discussion: A common answer is **hot weather** or **season**. Warm months bring more ice cream purchases and more swimming or water exposure, which can increase drownings.

Controlled Experiments

- Controlled experiments compare an **experimental group** with a **control (baseline) group**.
- Ideally, the groups are similar in all important respects except for the **factor being changed**, or **predictor variable (x)**.

Emphasize: We want x to be the only systematic difference between the groups.

- If the groups are otherwise similar, a difference in the **response variable (y)** can reasonably be attributed to the change in x .

Example: Compare the strength of parts manufactured using two different heat-treatment conditions while keeping material, geometry, and testing procedure the same.

- If the groups also differ in other important ways, these factors may be **confounded** with the predictor variable.

Ask: If one set of parts is printed on a newer machine and also uses a different process

Randomized Experiments in Engineering

- **Randomization** is often used to reduce confounding and systematic bias.
Connect back: Randomization helps prevent other variables from consistently favoring one experimental condition.
- Experimental units—such as parts, samples, devices, or test runs—are randomly assigned to the **experimental condition** or **control condition**.
Example: Randomly assign manufactured specimens to two different coating processes rather than testing one process first and the other process later.
- Randomization helps distribute both known and unknown sources of variation across the groups.
Ask: What could go wrong if all samples for Process A are tested in the morning and all samples for Process B are tested in the afternoon? Possible confounding factors: temperature, humidity, machine drift, operator, material batch, or equipment calibration.
- Good experimental design combines **control**, **randomization**, and **consistent measurement procedures** to isolate the effect of x on y .

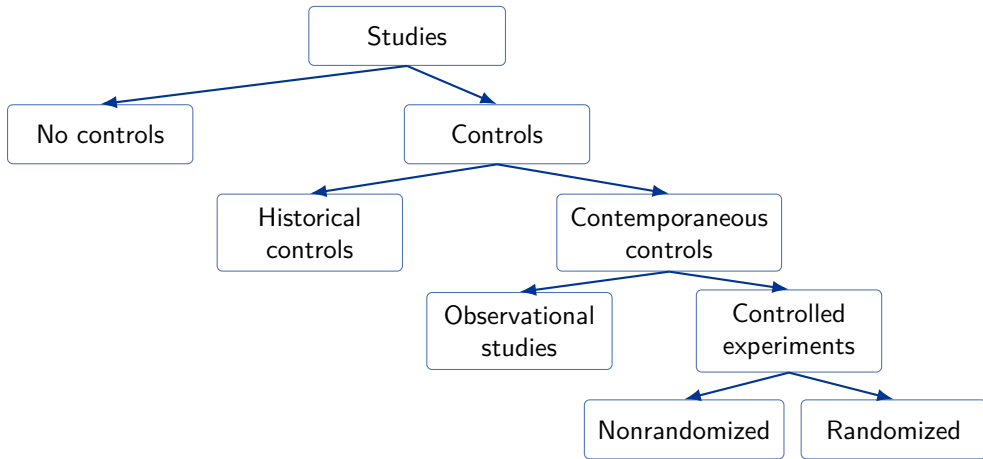
Medical Studies

- Subjects are randomly assigned to the **treatment group** or **control group**. Randomization helps make the groups similar with respect to other factors.
Ask: Why random assignment rather than letting patients choose which group they want to join?
Discuss: With a sufficiently large sample, randomization tends to balance both known and unknown factors across the groups.
- When possible, the control group receives a **placebo**: an inactive treatment that resembles the actual treatment.
Example: A sugar pill that looks identical to the actual medication.
- A placebo helps separate the effect of the treatment from the effect of *believing* that a treatment was received.
Ask: Can simply believing that you received a treatment affect the response? This is the placebo effect.

Observational Studies

- In an **observational study**, investigators do not assign subjects to treatment or control groups. Instead, they observe groups that differ naturally in the treatment, exposure, or condition of interest.
- Observational studies can identify **associations**: changes in one variable are related to changes in another.
- An association may suggest a possible causal relationship, but **association alone does not establish causation**.
- A major challenge is **confounding**. A third variable may be related to both the treatment or exposure and the response, making it difficult to determine what actually caused the observed difference.
- Confounding can sometimes be reduced by comparing groups that are similar with respect to important confounding variables.

Design of Experiments



Was there any control group at all? Were historical controls used, or contemporaneous controls? How were subjects assigned to treatment-through a process under the control of

Salk Vaccine: Two Designs

National Foundation for Infantile Paralysis (NFIP) design

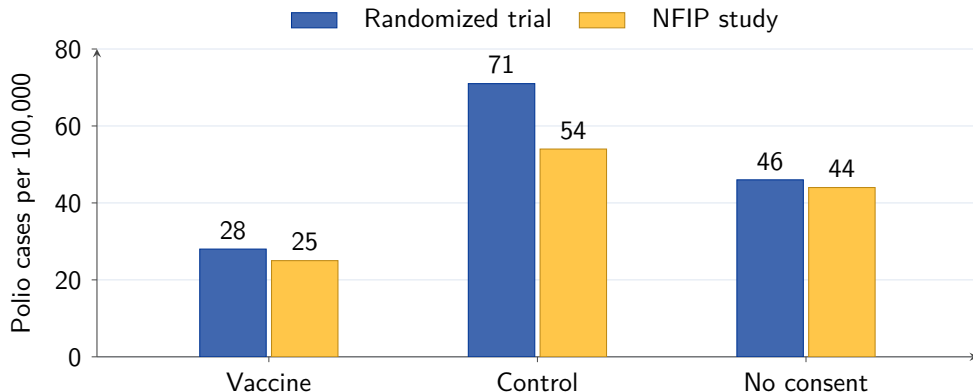
- Vaccinate grade 2 children whose parents consented.
- Use grades 1 and 3 as controls.
- No placebo or blinding.

Randomized controlled double-blind design

- Start with children whose parents consented.
- Randomly assign vaccine or placebo.
- Keep children and diagnosticians blind to assignment.

Which design was better? **Second. grade, exposure, and parental consent were mixed up with treatment. treatment and control groups were made comparable by chance. Source of confounding. higher-income parents were more likely to consent. Higher-income children more vulnerable to polio. better sanitation delayed their first exposure to poliovirus. In poorer or less sanitary environments, poliovirus circulated widely. Many children were infected very early in infancy, when they still had some protection from maternal antibodies**

Salk Vaccine Trial Results



Confounding in the NFIP study. Biased against vaccine. The NFIP treatment group included only children whose parents consented to vaccination. However, the control group also included children whose parents would not have consented. The control group was not comparable to the treatment group.

Ultrasound and Birthweight

Observed association

In an observational study, babies whose mothers received ultrasound examinations had **lower birthweight**, on average.

The association remained even after adjusting for several measured variables.

Does this mean ultrasound causes lower birthweight?

Why might patients who received ultrasound differ systematically from those who did not?

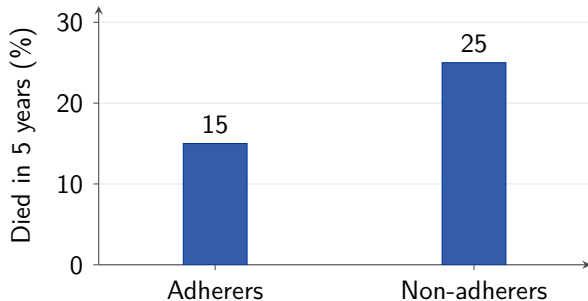
Problem pregnancy is a confounder:

Discussion: Ultrasound may be used more often in problem pregnancies, which are also more likely to result in low birthweight. Statistical adjustment can reduce confounding, but it does not guarantee that all confounding has been removed.

Historical Controls

- A **historical control** is a control group taken from past data rather than from subjects observed at the same time as the treatment group.
- **Medical example:** A hospital evaluates a new surgical procedure by comparing current patients with patients who received the previous procedure several years ago.
- **Engineering example:** A manufacturer compares the failure rate of a redesigned component with historical failure data from the previous design.
- **Problem:** Other conditions may have changed over time, so differences between the groups may be **confounded with time**.

Clofibrate: Does Taking the Drug Help?

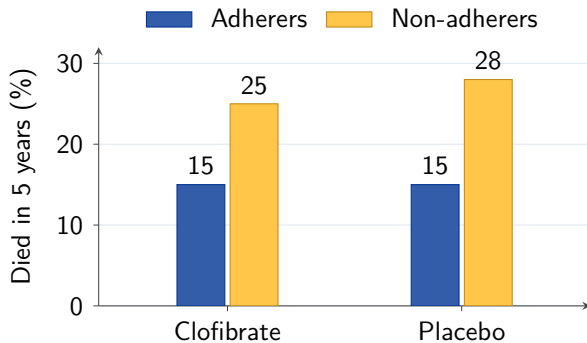


Among patients assigned clofibrate

Patients who took the drug regularly had substantially lower mortality than those who did not.

Does this show that taking clofibrate reduced mortality?

Clofibrate: Why Do We Need a Control Group?



What does the placebo group show?

Placebo adherers also had much lower mortality than placebo non-adherers.

The same pattern appears even when the “treatment” is a placebo.

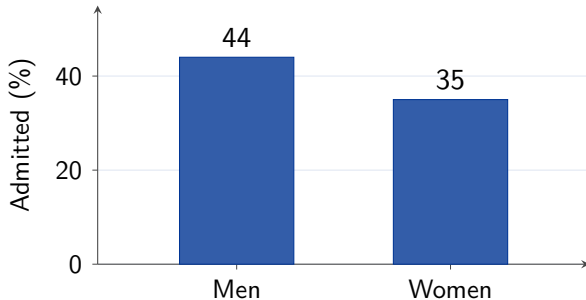
So adherence itself is associated with mortality.

Adherers may have differed from non-adherers in other ways—for example, health behaviors, overall health, or ability to follow medical advice. The control group reveals that the lower mortality cannot simply be attributed to clofibrate.

Clofibrate table and interpretation from Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), Ch. 2, pp. 13–14, Table 1..

Berkeley Graduate Admissions

What could differ between male and female applicants besides sex?



Question

Does this prove that the University discriminated against women?

Discussion: One key factor was choice of major/department. Applicants did not apply to the same mix of departments, and departments differed greatly in admission rates.

Berkeley Admissions: Control for Department

Major	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

Look across rows. Do you still see a large disadvantage for women within departments?

What changed after controlling for department?

Within the largest departments, male and female admission rates were much closer. Some departments favored women; others favored men.

Simpson's Paradox: How the Reversal Happened

Different application patterns

- Departments A and B were much easier to enter.
- Over half of male applicants applied to A or B.
- Over 90% of female applicants applied to the harder departments C–F.

Adjust for department

Imagine that men and women applied to the six departments in **exactly the same proportions**.

Use the same department mix to average the admission rates for each group:

Men: 39% Women: 43%

Why did the comparison reverse?

The overall rates mixed together departments with very different admission rates. Men applied more often to easier departments, while women applied more often to harder departments.

**After controlling for department,
the apparent overall advantage for men disappears.**

Summary

- **Association alone does not establish causation.** A confounder may explain an observed relationship.
- **Randomized experiments provide stronger causal evidence.** Random assignment helps balance confounders. Blinding and placebos can reduce bias.
- **Observational studies require careful comparisons.** Groups form through choice or circumstance. Adjustment can reduce confounding, but important differences may remain.
- **Overall and subgroup comparisons can differ.** Simpson's paradox shows that an association can reverse when groups are combined.